

has only the trivial solution. The set $\{v_1, \dots, v_p\}$ is said to be linearly dependent if there exists weights $c_1, \dots, c_p$, not all zero, such that

$$c_1 v_1 + c_2 v_2 + \dots + c_p v_p = 0 \quad (2) \quad \square$$

Equation (2) is called a linear dependence relation among $v_1, \dots, v_p$ when the weights are not all zero.

Example 1: Let $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$ and $v_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ - independent.

- Determine if the set $\{v_1, v_2, v_3\}$ is linearly independent.
- If possible, find a linear dependence relation among v_1, v_2 and v_3 .

Solution: a) $x_1 v_1 + x_2 v_2 + x_3 v_3 = 0 \Rightarrow x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

By Defn. of Matrix eqn. $Ax=b$, we get:

$$\begin{bmatrix} 1 & 4 & 2 \\ 2 & 5 & 1 \\ 3 & 6 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From Theorem 3, the above matrix equation has the same solution set as the system of eqns. corresponding to the augmented matrix

$$\begin{bmatrix} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{bmatrix}$$

This system of eqns. has the unique trivial solution if there are no free variables (Theorem 2)

- By defn, a free variable corresponds to ~~the~~ a column having no pivots
- By defn, a pivot position corresponds to a leading 1 of a row in ref.

$$\begin{aligned} & \begin{bmatrix} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{bmatrix} \xrightarrow[\text{Row 2} = \text{Row 2} - 2 \times \text{Row 1}]{\text{Row 3} = \text{Row 3} - 3 \times \text{Row 1}} \begin{bmatrix} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & -6 & -6 & 0 \end{bmatrix} \\ & \xrightarrow[\text{Row 3} = \text{Row 3} - 2 \times \text{Row 2}]{\text{Row 2} = \text{Row 2} \times (-1/3)} \begin{bmatrix} 1 & 4 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{Row 1} = \text{Row 1} - 4 \times \text{Row 2}]{\text{Row 1} = \text{Row 1} - 4 \times \text{Row 2}} \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$